GENESIS: A Homeostatic-Reward-Only Artificial Ecology for Studying What Social Structure Can Emerge, with First Results from Recurrent PPO Agents

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GENESIS project

https://github.com/HeetRanpura/Genesis

Abstract

Most multi-agent reinforcement learning (MARL) studies of cooperation build part of the answer into the reward: team rewards, prosocial bonuses, influence terms or communication losses. Under those designs it is hard to say whether a social behaviour *emerged* or was *specified*. We present GENESIS, a seeded, procedurally generated foraging ecology in which every agent is rewarded *only* for keeping its own body near a physiological setpoint (homeostatic reinforcement learning), plus a death penalty. The rule is enforced by a unit test. Ecological features that make social behaviour possible or useful, but never rewarding, are documented as *affordances*: large prey that need two simultaneous hunters, food that can be carried and dropped, scarce fresh water, and seasonal winters. Every cognitive or social capacity is a configuration switch that is off by default. The engine is a fused Numba kernel. It simulates a 256×256 world with 1,000 agents, three species of food-limited scripted fauna, weather, fire and floods at ≈112k steps per minute including observations, on a laptop CPU. Runs are bit-reproducible, and every agent at every tick is recorded and exactly replayable. We pair the engine with a preregistered evaluation protocol: training on worlds 1000+, evaluation on held-out worlds 0–4, three non-learning baselines in every comparison, and bootstrap confidence intervals with Hedges' g.

In a first study, parameter-shared recurrent (GRU) and memoryless PPO agents trained for 19.7M agent-steps reach 1.80 and 1.58 deaths / agent-yr on held-out worlds. Both clearly beat random (3.83) and greedy (3.31) foragers but cannot be distinguished from a hand-written heuristic (1.50). Decomposing mortality shows why. The learners nearly eliminate thirst, predator and fire deaths (thirst 0.09 vs. 0.96 deaths / agent-yr for the heuristic), but die of starvation, mostly in winter. Over the first half-year (spring–summer) their mortality is 4–6× lower than the heuristic's (0.19 and 0.33 vs. 1.22). In winter it is 1.4–1.9× higher. A first-order energy budget computed from the recorded event stream shows that the learners' intake (1.31–1.34×10^{-3} per tick, 73–74% from plants) barely covers basal metabolism (1.3×10^{-3}). The memoryless agent survives better largely because it moves less and rests more (48% of winter actions vs. 23% for the GRU agent). Memory reduced thirst deaths 2.7-fold but did not reduce all-cause mortality at this budget. Learners also persist in futile behaviour: 78–91% of their DRINK actions during droughts happen with no water in reach, against 30–37% otherwise. We report these results as pilots, with all negative results kept. We describe the measurement framework, null models and preregistered replication (3 seeds per arm, in progress) that will test whether any social organisation appears at all. Applying the preregistered internal-state and social analyses to the pilot checkpoints, we find four things. Linear probes decode own hydration, energy and predator presence from the GRU state (R^2 0.70 and 0.85; AUC 0.91). The value estimate predicts death within 200 ticks (AUC 0.98). The GRU state carries no more information about out-of-view water than the network's current encoding does. Learners cluster more than water and habitat explain, but blinding them to other agents leaves the clustering intact, so it is not social. We found no cooperative large-prey hunting.

Preprint. Work in progress; numbers are from pilot runs (one training seed per arm). 27 October 2026.

1 Introduction

Human social life (sharing food, hunting together, signalling, keeping traditions, enforcing norms) appears in the archaeological and ethnographic record in a rough order, under particular ecological conditions (Tennie et al., 2009; Tomasello, 1999). A computational testbed could probe the *structural* side of this question. Given learners that care only about their own bodies, and an ecology in which cooperation is possible and sometimes useful, which social regularities appear, in what order, and under which conditions? We are explicit about the scope of the analogy. Our agents are small neural policies, not models of early humans, and any comparison we eventually draw is between orders of emergence and triggering conditions, not between minds.

The confound we want to remove. Much MARL research on cooperation adds social terms to the objective. Examples are inequity aversion (Hughes et al., 2018), social-influence bonuses (Jaques et al., 2019), shared team rewards, or differentiable communication channels trained end to end (Foerster et al., 2016). These are excellent tools for *engineering* cooperation. When the question is whether cooperation *emerges*, they make the answer partly true by construction. Benchmarks such as Melting Pot (Leibo et al., 2021) and sequential social dilemmas (Leibo et al., 2017) instead fix a game and ask how well agents play it. Large persistent worlds such as Neural MMO (Suarez et al., 2019) provide scale and competition but are not built around physiological need. Artificial-life systems such as Tierra and PolyWorld (Ray, 1991; Yaeger, 1994) pioneered open-ended ecologies, mostly without modern deep RL learners or preregistered statistics.

Our approach. Genesis combines four commitments:

1. **Reward purity.** Each agent's reward is the reduction of its own homeostatic drive, following Keramati and Gutkin (2014), plus a death penalty. The reward has no social term, shaping or curiosity bonus. A unit test fails if the reward ever depends on anything but the agent's own body.
2. **Affordances, not incentives.** Every ecological feature that could make social behaviour useful (cooperatively hunted large prey, droppable food, a kin-similarity cue, optional huddling warmth) is tagged and switchable, so its effect can be ablated.
3. **Capabilities as switches.** Memory, world models, communication, social learning, reputation, niche construction, reproduction and aggression are all off by default. Experiments turn on exactly one thing at a time.
4. **Evaluation discipline.** Agents train and are evaluated on disjoint world seeds. Three non-learning baselines appear in every comparison. Hypotheses and decision rules are preregistered before evaluation (Nosek et al., 2018), and uncertainty is reported over seeds (Agarwal et al., 2021; Henderson et al., 2018).

Fig. 1 gives the overall architecture.

Contributions.

- **An engine** (§3). A deterministic, food-limited, seasonal ecology with 1,000+ agents at ≈112k steps/min with observations on a laptop CPU. It has a counter-based RNG, so results do not depend on the thread count, an incrementally maintained observation image, precomputed line-of-sight masks, and a flight recorder that makes every run exactly replayable.
- **A protocol** (§5). Held-out worlds, baselines, preregistration, a strict non-overlap rule for "beats", and explicit units of replication.
- **First learner results** (§6). Survival-only PPO agents beat random and greedy foragers but not a hand-written heuristic. A season-resolved, cause-resolved and energy-resolved analysis attributes the gap to a winter energy deficit. Memory trades thirst deaths for starvation deaths, and learned policies remain inflexible when the environment changes (droughts). Probes, value calibration,

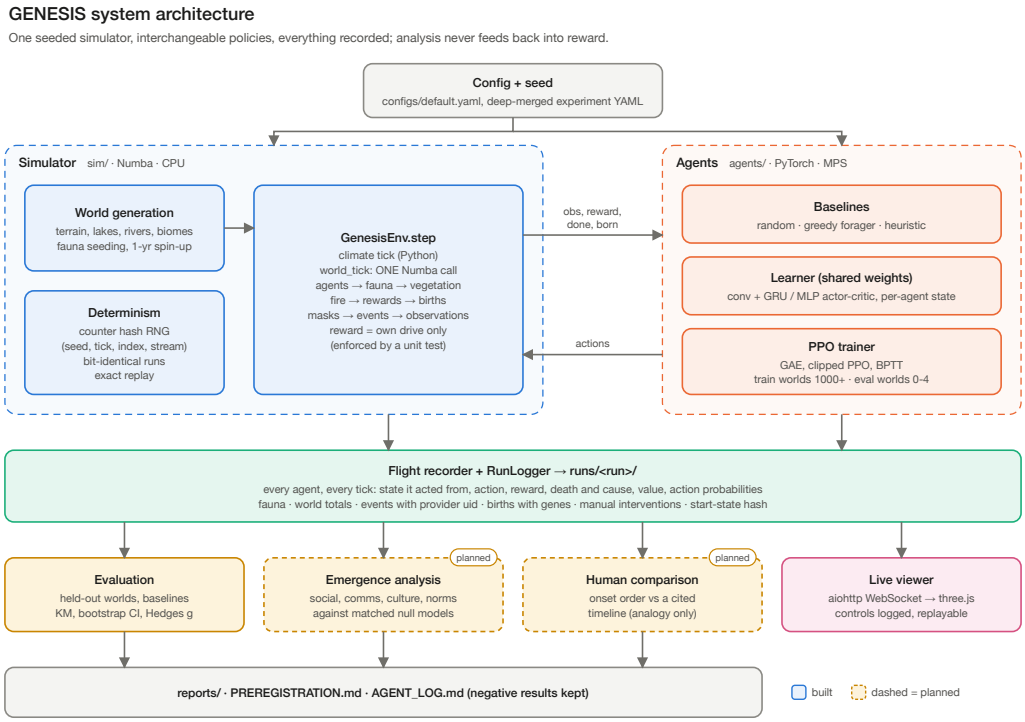


Figure 1: System architecture. A seeded simulator and interchangeable policies exchange observations and actions; a flight recorder captures every agent at every tick. Evaluation, emergence analysis and the viewer only read the recordings, so nothing downstream can feed back into the reward. Dashed boxes are planned.

null models and a causal "social blinding" ablation separate genuine social structure from look-alikes (§6.7).

- **A measurement framework** (§7). Operational definitions, null models and planted-truth unit tests for social, communicative and cultural phenomena, preregistered before any of them is measured.

We stress what we do *not* claim. No social behaviour has yet been measured with the Stage-3 instruments, and we expect none in the capability-off setting reported here (preregistered H4/H7).

2 Related work

Homeostatic and need-based RL. Keramati and Gutkin (2014) showed that rewarding the reduction of a drive, defined as a distance from a physiological setpoint, makes reward seeking and physiological stability the same objective. They also derived temporal discounting as a pressure toward the shortest path to the setpoint. We adopt their drive-reduction reward verbatim over four variables: energy, hydration, health and warmth.

Emergent cooperation and social learning in MARL. Sequential social dilemmas (Leibo et al., 2017) and the Melting Pot suite (Leibo et al., 2021) evaluate cooperation and generalisation to new co-players. Emergent tool use (Baker et al., 2020), bartering (Johanson et al., 2022) and social learning (Bhoopchand et al., 2023; Ndousse et al., 2021) show that rich behaviour can arise from simple objectives when the environment supports it. Ndousse et al. (2021) in particular found that independent model-free agents rarely learn from experts without auxiliary losses. This is a warning for our capability-off agents. Parameter-shared PPO is a strong default in cooperative MARL (Yu et al.,

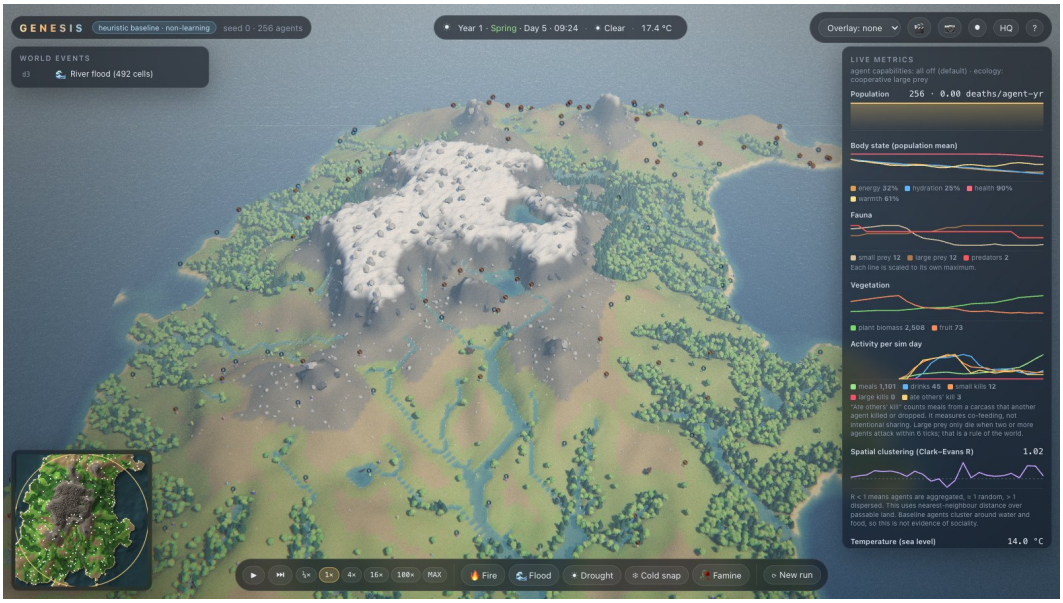


Figure 2: The live viewer on held-out world 0 (three.js; §D). Agents (dots) forage across forest, grassland and river networks around a central mountain massif. The right panel shows live, measured quantities: population, mean body state, fauna, vegetation, activity and the Clark–Evans aggregation index. Terrain relief and foliage are visual only. Everything on the dashboard is computed from simulator state.

2022).

Emergent communication. Agents can learn referential signalling games (Lazaridou et al., 2017) and differentiable channels (Foerster et al., 2016). Lowe et al. (2019) showed that common metrics can mislead, and recommend measuring both positive signalling and positive listening, with causal tests. We adopt their distinction.

Artificial life and evolution. Tierra (Ray, 1991) and PolyWorld (Yaeger, 1994) pioneered open-ended ecologies. The Baldwin effect (Hinton and Nowlan, 1987) motivates our planned learning-plus-evolution experiments.

Continual learning. Our agents live in a non-stationary world (seasons, disturbances, new worlds). Catastrophic forgetting (Kirkpatrick et al., 2017) and loss of plasticity (Dohare et al., 2024) are therefore first-class concerns, not afterthoughts.

Cultural evolution. We operationalise traditions following the exclusion method of Whiten et al. (1999), the cumulative ratchet following Tennie et al. (2009), and communication properties following Hockett's design features (Hockett, 1960).

3 The Genesis world

3.1 Time, terrain and water

One tick is one action per agent. A day is 120 ticks, a season 8 days, and a year 3,840 ticks: spring, summer, autumn, winter. Every run begins with one simulated year of ecology without agents (*spin-up*), so the lush initial state's boom and crash is not part of the experiment.

The world (default 256×256) is generated from the seed in these steps:

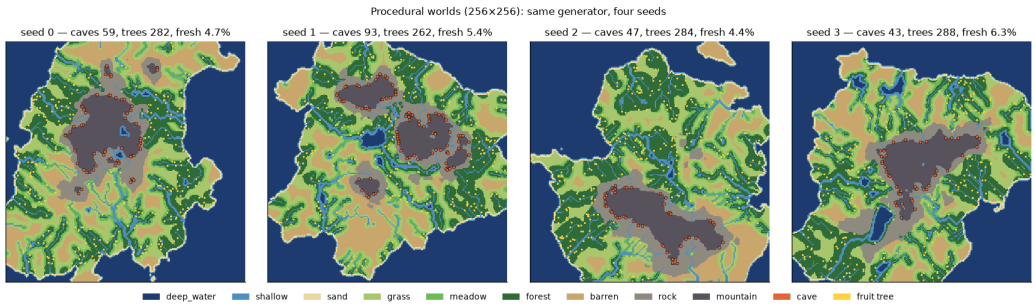


Figure 3: Procedural worlds from four seeds. Fresh water covers 4.4–6.3% of the land. Caves (red) are shelter; fruit trees are yellow. Seeds 0–4 are reserved for evaluation and never used in training.

1. domain-warped fractal elevation with a continental falloff, then rank normalisation;
2. priority-flood depression filling (Barnes et al., 2014), which turns filled depressions into lakes;
3. D8 flow accumulation (O'Callaghan and Mark, 1984): rivers at ≥180 upstream cells, streams at ≥90;
4. a moisture field;
5. biomes: deep and shallow water, sand, grass, meadow, forest, barren, rock, mountain and caves.

The ocean is *salt*. Fresh water comes only from rivers, streams and lakes, which cover about 4–6% of the map (Fig. 3). Water is therefore spatially scarce and often out of view. That is what can make memory useful.

3.2 Resources and fauna

Plants. Plants regrow logistically toward a per-cell fertility. Growth is scaled by season (spring 1.2, summer 1.0, autumn 0.55, winter 0.05), by rain (+25%), drought (×0.4), famine (×0.25) and temperature (none below 2 °C). A grazed cell regrows from a 10% floor (roots). One EAT bite removes 0.25 biomass, worth at most 0.03 energy.

Fruit trees. Each tree fruits in one season (up to 6 fruit at 0.15 energy each).

Carcasses. Carcasses spoil linearly. Each carcass remembers the uid of whoever killed or dropped it, so every meal records its provider, which is what the sharing analysis needs.

Fauna. The three animal species are scripted, because they are part of the environment, and *food-limited*: each animal has an energy store, a metabolism (torpor at ×0.2 below 5 °C), grazing satiation, and seasonal births from a full store.

- **Small prey** flee within 4 cells. One hunter suffices: a kill succeeds with probability 0.6 and leaves 0.30 energy.
- **Large prey** also browse woody forage in forests. Without this niche, small prey competitively exclude them.
- **Predators** hunt agents and prey within 8 cells and never enter caves.

Population caps and immigration floors exist only as safety rails. Over 5 seeds × 3 years the caps never bind (0% of time), and the floors bind rarely. The exception is predators, which sit at their floor about 31% of the time when 256 heuristic agents compete with them for small prey. We report this as a known weakness.

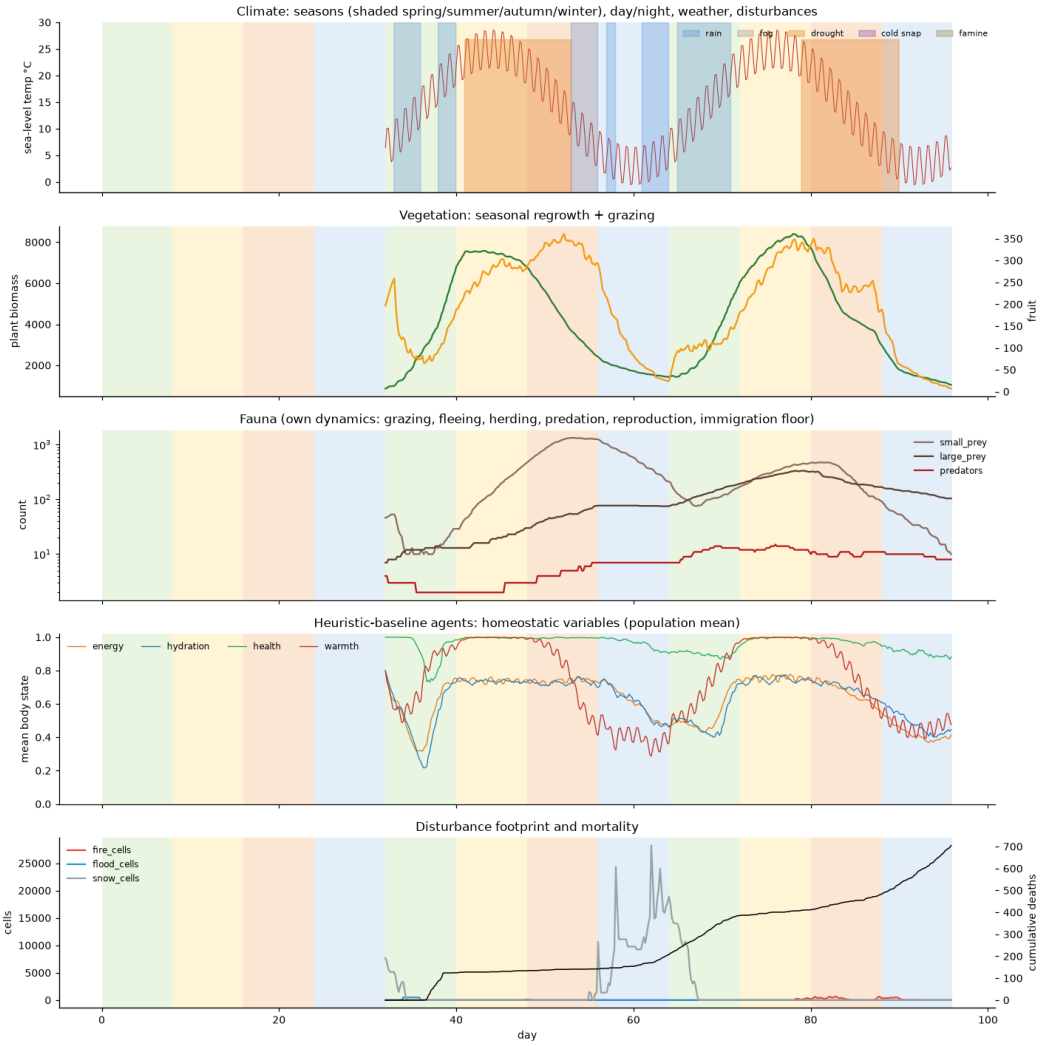


Figure 4: Two years of one world (seed 0, heuristic agents). From top: sea-level temperature with weather and disturbance episodes; plant biomass and fruit; food-limited fauna (log scale); population-mean body state; disturbance footprint and cumulative deaths. Shading marks the seasons. Spin-up (the first 32 days) is excluded from all statistics.

3.3 Weather and disturbances

A daily Markov chain drives rain, fog (view radius 3), summer droughts (streams dry up; plant growth $\times 0.4$; fire risk $\times 4$), winter cold snaps ($-10\,^{\circ}\mathrm{C}$) and famine years. Wildfires ignite daily with a small probability, spread by fuel and dryness, are put out by rain and leave burn scars. Spring floods turn floodplains into shallow water. Snow builds up where it rains below $0\,^{\circ}\mathrm{C}$. All disturbances can also be triggered manually; every trigger is logged with all its parameters and replayed exactly (Fig. 4).

3.4 Agents: body, perception, actions

Body. Each agent's body has four homeostatic variables $\boldsymbol{h} = (e, w, \eta, \theta) \in [0, 1]^4$: energy, hydration, health and warmth. It also has an age, and a heritable lifespan of 3–4 years. Energy falls by a basal cost of 1.3×10^{-3} per tick, halved while resting and scaled by body size, plus 9×10^{-4} per move, scaled by terrain ($\times 1.25$ forest, $\times 1.6$ rock, $\times 2$ shallow water, more in snow), plus a carrying cost.

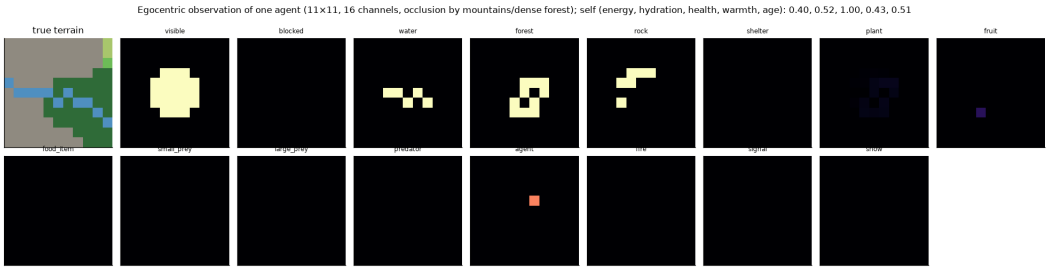


Figure 5: One agent's observation. True terrain (left) and the 16 egocentric channels after line-of-sight masking. Two forest cells or a mountain block sight. The agent sees only the bright disc in "visible".

- **Hydration** falls by 1.6×10^{-3} per tick (×2 above 31 °C, ×0.7 while resting).
- **Warmth** relaxes toward a target set by the effective temperature, which depends on altitude, shelter, forest, rain, snow and resting.
- **Health** falls when energy or hydration hits zero, when warmth is below 0.2, and from fire, predators, large-prey kicks or strikes. It regenerates when all needs are met.

Genes. Eight floats: lifespan, body size, cold tolerance and five neutral markers. The markers drive appearance and the [affordance] kin-similarity cue.

Perception.

- **Grid view.** An egocentric $11 \times 11 \times 16$ uint8 grid (Fig. 5). The 16 channels are: visible, blocked, water, forest, rock, shelter, plant, fruit, food item, small prey, large prey, predator, agent (value encodes genetic similarity), fire, signal and snow.
- **Occlusion.** Line of sight is blocked by mountains and by a second forest cell. The view radius shrinks from 5 to 2 at night and to 3 in fog.
- **Vector inputs.** A 13-feature interoceptive self-vector, up to 4 visible neighbours (9 features each), and heard signals (used only when communication is on).

Actions. There are 11 base actions: noop, move N/E/S/W, rest, eat, drink, hunt, pickup and drop. Switches add build, strike and n_{sym} signal actions.

- **Drink** succeeds only next to fresh water or floodwater, or anywhere while it rains (a small amount).
- **Food transfer.** Pickup and drop are always available, so food can be carried and dropped where another agent can take it. This is the [affordance] food-transfer affordance.

3.5 Reward (the hard rule)

With setpoint $\boldsymbol{h}^* = (1, 1, 1, 0.8)$, the drive is

$$D(\boldsymbol{h}) = \left\| \max(\boldsymbol{h}^* - \boldsymbol{h},\, 0) \right\|_2, \tag{1}$$

and the reward is

$$r_t = D(\boldsymbol{h}_t) - D(\boldsymbol{h}_{t+1}) + c_{\text{death}} \mathbb{1}[\text{died}_t] \left(+ \, c_{\text{repro}} \mathbb{1}[\text{reproduced}_t] \right), \tag{2}$$

with $c_{\text{death}} = -1$ and $c_{\text{repro}} = 0$ by default. Eq. 2 depends only on the agent's own body. The test `test_reward_is_own_homeostatic_drive_only` recomputes Eq. 2 from each agent's body state alone over 200 random-action ticks and asserts that it equals the environment's reward exactly.

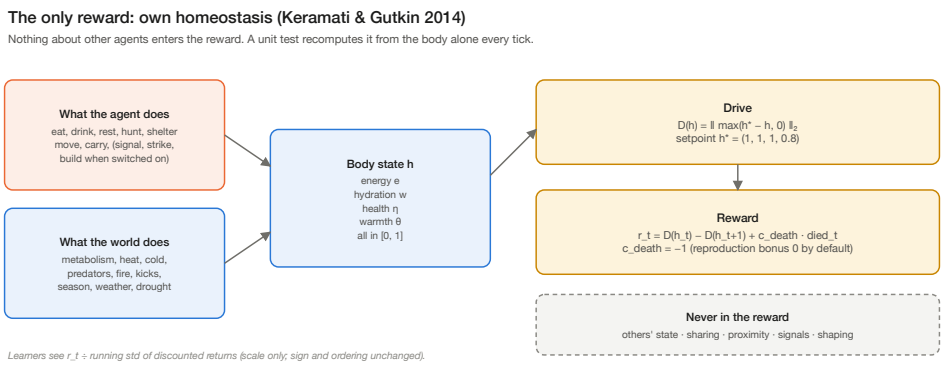


Figure 6: The reward. Actions and the world change the body; the reward is the reduction of the agent's own drive plus a death penalty. Nothing about other agents enters it.

Table 1: Switches. Affordances (A) change the ecology; capabilities (C) change what agents can do or perceive. Defaults are in the right column.

Switch	Type	Effect	Default
`cooperative_large_prey`	A	large-prey kill needs $\geq$ 2 distinct attackers within 6 ticks; carcass 1.8 energy	on
`huddle_warmth`	A	each adjacent agent adds +1.5 °C (biases aggregation)	off
food pickup/drop	A	food can be carried and left for others; provider logged	always
agent channel value	A	encodes genetic similarity of visible agents (kin cue)	always
`memory`	C	recurrent core (GRU) in the policy	off
`world_model`	C	latent dynamics model and imagination (planned)	off
`communication`	C	n_{sym}=8 costly signals, heard within 9 cells	off
`social_learning`	C	neighbours' last actions visible	off
`reputation`	C	neighbours' share/strike histories visible	off
`aggression`	C	STRIKE action (makes punishment possible)	off
`niche_construction`	C	wood/stone pickup, BUILD shelters	off
`reproduction`	C	births with mutated heritable genes; no respawn	off

3.6 Affordances and capability switches

Table 1 lists every switch. Affordances make social behaviour *possible or useful*; nothing makes it *rewarding*. In the experiments reported here, all capabilities are off and only cooperative large prey is on.

3.7 Engine design and performance

One tick is a Python climate update followed by *one* Numba (Lam et al., 2015) call, `world_tick`. It runs agents, fauna, vegetation and trees, fire, rewards, births and respawns, reward masks, event tally and observations in order. Figs. 21–23 (Appendix) detail generation, the tick and observations. Three design choices make it fast and exact:

1. **Counter-based RNG.** All in-kernel randomness is a SplitMix-style hash of (seed, tick, index, stream) (Salmon et al., 2011; Steele et al., 2014). Parallel loops draw no shared state, and sequential resolution follows hash-derived permutations. A run is therefore a pure function of (config, seed, actions), whatever the thread count.
2. **Incremental observation image.** Observations are masked crops of a world image $I[y, x, 16]$ that is patched in place at every mutation site, not rebuilt every tick. A test compares it with a

Table 2: Engine throughput (median of interleaved runs; Apple M4 laptop CPU, fanless, so absolute numbers are conservative). "Obs" includes building every agent's observation.

World	Agents	Obs	Steps/min	Agent-steps/s
256^2	256	no	211,287	0.90M
256^2	256	yes	146,188	0.62M
256^2	1,000	no	171,568	2.86M
256^2	1,000	yes	**111,992**	**1.87M**
256^2	2,000	yes	89,041	2.97M
512^2	1,000	yes	59,499	0.99M

full rebuild.

3. **Static line of sight.** Terrain never changes, so each cell's line-of-sight mask is precomputed once by propagation along Bresenham parent chains.

Table 2 reports throughput. A PettingZoo `ParallelEnv` wrapper (Terry et al., 2021) passes the official API test.

4 Agents and training

4.1 Baselines

Three non-learning policies act on exactly the observations learners receive. None has a social rule.

- **Random** acts uniformly over the actions.
- **Greedy forager** attends to its lower of energy and hydration. It eats or drinks if possible, otherwise walks to the nearest visible resource for that need, otherwise explores.
- **Heuristic** follows hand-written survival rules. It flees predators and fire. It drinks below hydration 0.55 until 0.95 and eats below energy 0.6 until 0.9 (with hysteresis). It also hunts adjacent small prey, seeks shelter when cold and rests at night when sated.

4.2 Learner

Architecture. All agents share one actor-critic (about 1.1M parameters), and each keeps its own recurrent state.

- **Grid branch:** conv3×3(32), then conv3×3 stride 2 (64), then FC 256.
- **Vector branch:** FC 64 over the self, neighbour and signal inputs.
- **Trunk and heads:** the two branches are concatenated, passed through FC 256 and then the core, then policy and value heads.
- **Core:** `GRUCell`(256) (Cho et al., 2014) in the *GRU* arm; a 256-unit feed-forward layer in the *memoryless* arm, otherwise identical.

A slot's hidden state is zeroed when a new agent occupies it.

Learning rule. Clipped PPO (Schulman et al., 2017) with GAE (Schulman et al., 2016). Fig. 7 shows the network, and Fig. 8, Algorithm 1 and Table 3 give the loop and hyper-parameters.

- Each of the 256 slots is a continuing experience stream.
- A death cuts the bootstrap, because the slot's next observation belongs to a newborn.

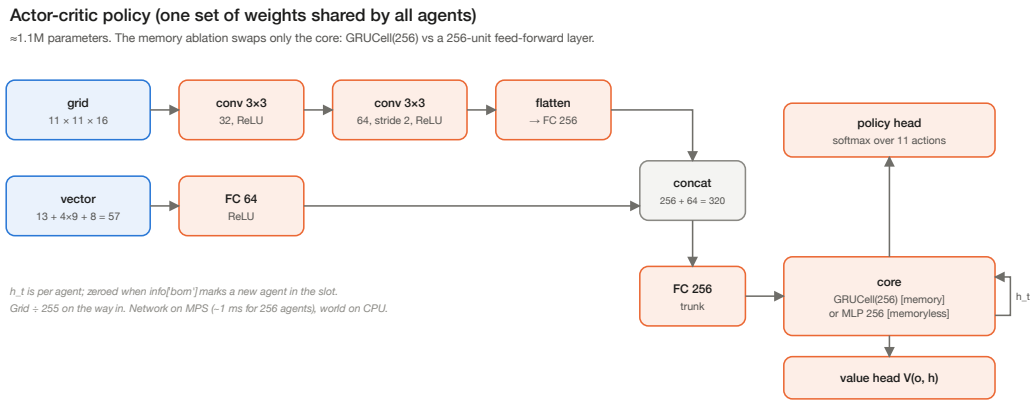


Figure 7: Policy network. One set of weights is shared by all agents; each agent has its own recurrent state. The memoryless arm replaces only the GRU core.

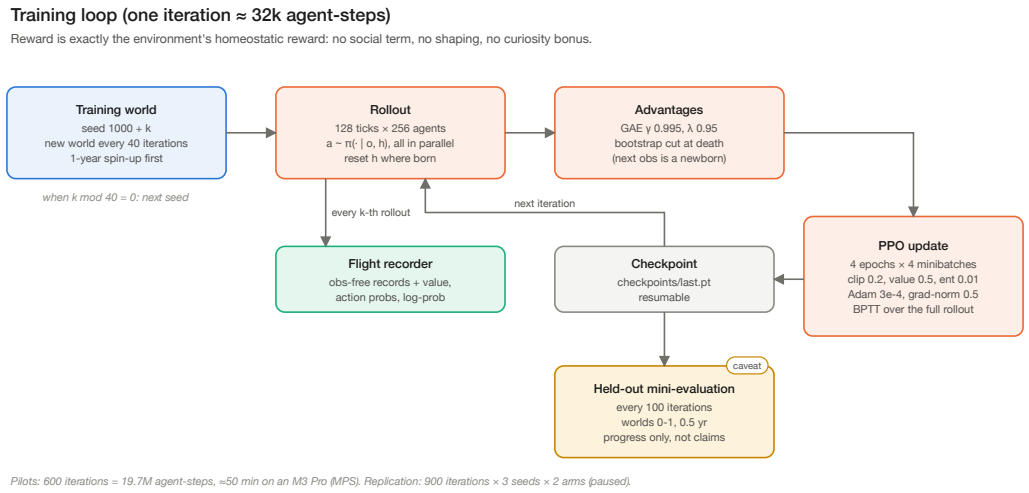


Figure 8: Training loop. Each iteration collects 128 ticks from 256 agents, computes GAE advantages with the bootstrap cut at deaths, and runs a clipped-PPO update. Held-out mini-evaluations use a half-year horizon and track progress only (see Fig. 12).

- The GRU arm is trained by back-propagation through the whole 128-tick rollout, starting from the hidden state stored at rollout time.

The training world changes every 40 iterations (seeds 1000, 1001, . . .), so agents see seasons, droughts and new maps. The network runs on Apple MPS and the world on CPU.

5 Evaluation protocol and preregistration

Protocol. Every policy (learner or baseline) is run on held-out worlds 0–4 for 2 simulated years with 256 agents and a fixed population (a dead agent's slot is refilled at a random land cell). Actions are sampled, not taken greedily.

Lifetimes. Lifetimes are measured from the end of spin-up. An earlier version measured them from tick 0, which halved measured mortality; that bug is fixed and covered by a regression test.

Algorithm 1 Training loop of the pilots (original `agents/rppo.py`, since folded into `agents/ppo.py` with the same defaults)

world ← GENERATE(seed = 1000); spin up 1 year; spawn 256 agents
for iteration $k = 0, \ldots, K-1$ **do**
  if k mod 40 = 0 and $k > 0$ **then** world ← GENERATE(seed = 1000 + $k/40$)
  end if
  for $t = 1..128$ **do** ▷ rollout, all agents in parallel
    $a_t \sim \pi_\phi(\cdot \mid o_t, h_t)$; $(o_{t+1}, r_t, d_t, \text{born}_t) \leftarrow \text{WORLDTICK}(a_t)$
    $h_{t+1}[\text{born}_t] \leftarrow 0$; record $(o_t, a_t, \log \pi, V, r_t, d_t)$ to the flight recorder
  end for
  $\hat{A} \leftarrow$ GAE(γ=0.995, λ=0.95), bootstrap cut at deaths
  for 4 epochs × 4 minibatches over agents **do** ▷ full-sequence BPTT
    minimise $-L^{\text{clip}}(\epsilon{=}0.2) + 0.5\, L^V - 0.01\, \mathcal{H}[\pi]$; clip $\|\nabla\| \le 0.5$
  end for
  if k mod 100 = 0 **then** held-out mini-evaluation (worlds 0–1, 0.5 yr)
  end if
end for

Table 3: Learner hyper-parameters (identical across arms except the core).

Rollout	128 ticks × 256 agents	Discount γ / GAE λ	0.995 / 0.95
Iterations (pilot)	600 (19.7M agent-steps)	PPO clip ϵ	0.2
Epochs × minibatches	4 × 4	Learning rate (Adam)	3×10^{-4}
Value coef. / entropy coef.	0.5 / 0.01	Gradient-norm clip	0.5
Hidden size	256	World switch	every 40 iterations
Device	Apple MPS (M3 Pro)	Wall time per run	≈50 min

Metrics.

- mortality (deaths per agent-year lived);
- Kaplan–Meier survival at one year (Kaplan and Meier, 1958);
- median lifetime;
- cause-specific death rates;
- per-season mortality and energy reserves.

Statistics. The unit of replication in the pilot is the evaluation world ($n = 5$). We report the mean and a 95% percentile bootstrap CI over worlds (Efron, 1979), plus Hedges' g (Hedges, 1981). "A beats B" means the two CIs do not overlap; overlapping CIs are reported as "not distinguishable at $n = 5$", never as a win.

Preregistration. Hypotheses H1–H4 were committed to `docs/PREREGISTRATION.md` at 21:20 IST on 26 Oct 2026, before any evaluation:

- **H1:** the GRU learner beats random and greedy.
- **H2:** the GRU learner beats the heuristic.
- **H3:** the GRU learner has fewer thirst deaths than the memoryless learner, with all-cause mortality no higher.
- **H4:** no measurable social behaviour.

Analyses not in the preregistration are labelled *exploratory* below. These include the seasonal decomposition, the energy budget, the half-year horizon and drought inflexibility.

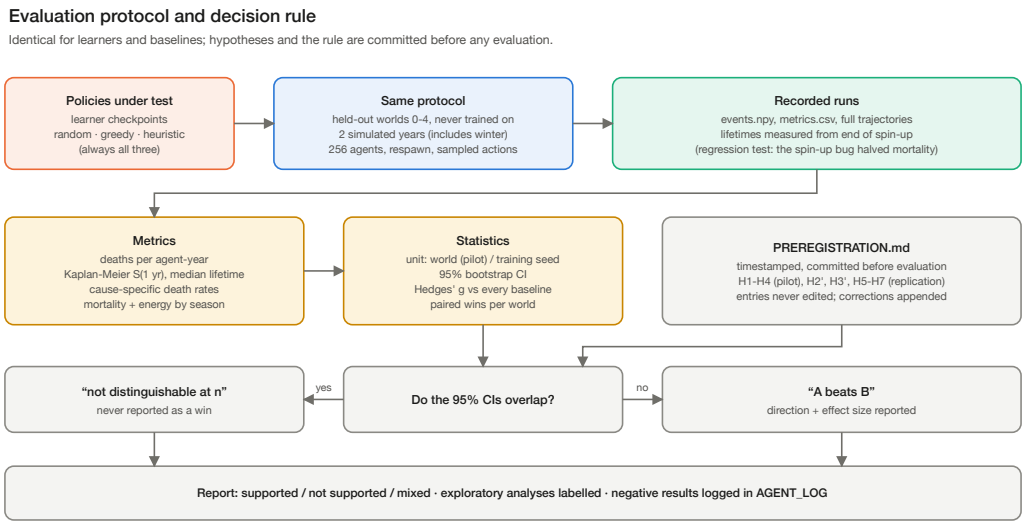


Figure 9: Evaluation protocol and decision rule. Learners and baselines run the same protocol on held-out worlds. "Beats" requires non-overlapping 95% CIs; the hypotheses and the rule are committed before evaluation.

Table 4: Held-out survival (worlds 0–4 × 2 yr, 256 agents). Mortality is the mean over worlds [95% bootstrap CI]. g is Hedges' g of the learner minus the row's baseline (negative = learner better). Wins (GRU, memoryless) count paired worlds on which the learner had lower mortality; $S(1\,\text{yr})$ is Kaplan–Meier survival at one year; life is the median lifetime in days.

Policy	deaths / agent-yr	$S(1\,\text{yr})$	life (d)	g GRU	g mem.	wins
Random	3.83 [3.53, 4.19]	0.001	6.5	−4.58	−5.54	5/5, 5/5
Greedy	3.31 [2.92, 3.77]	0.006	4.7	−2.97	−3.63	5/5, 5/5
Heuristic	1.50 [1.04, 2.11]	0.214	13.3	+0.50	+0.15	1/5, 1/5
GRU learner	1.80 [1.52, 2.10]	0.062	11.7	–		
Memoryless learner	1.58 [1.35, 1.83]	0.101	13.7		–	

6 Results

6.1 Learners beat naive foragers but not hand-written rules

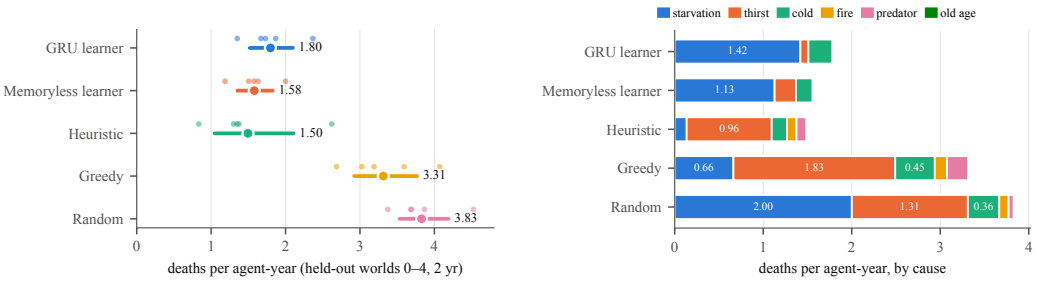
Table 4 and Fig. 10 give the primary outcome.

- **H1 is supported.** Both learners beat random and greedy on every world, with non-overlapping CIs and $|g| > 2.9$.
- **H2 is not supported.** Both learners' CIs overlap the heuristic's, and both point estimates are slightly worse. They beat it on only one of five worlds (world 3).

The two learners and the heuristic reach similar totals by completely different routes (Fig. 10b):

- **Thirst:** 0.09 [0.04, 0.14] (GRU) and 0.24 [0.11, 0.36] (memoryless), against 0.96 [0.62, 1.31] for the heuristic.
- **Predators and fire:** 0.006–0.011 for the learners, an order of magnitude below the heuristic's 0.11 each.
- **Starvation:** 1.42 [1.17, 1.62] and 1.13 [0.96, 1.28] for the learners, against 0.14 for the heuristic.

A survival-only learner with no water-seeking rule thus learned to manage hydration far better than our hand-written rules, and failed on energy.



(a) mortality, mean and 95% CI; small dots are worlds

(b) cause-specific death rates

Figure 10: Held-out survival over two years. (a) Both learners separate from random and greedy, with no overlap between CIs, but overlap the heuristic. (b) The causes differ completely. The heuristic dies of *thirst* (0.96 deaths / agent-yr). The learners die of *starvation* (1.42 and 1.13) and have almost no predator or fire deaths.

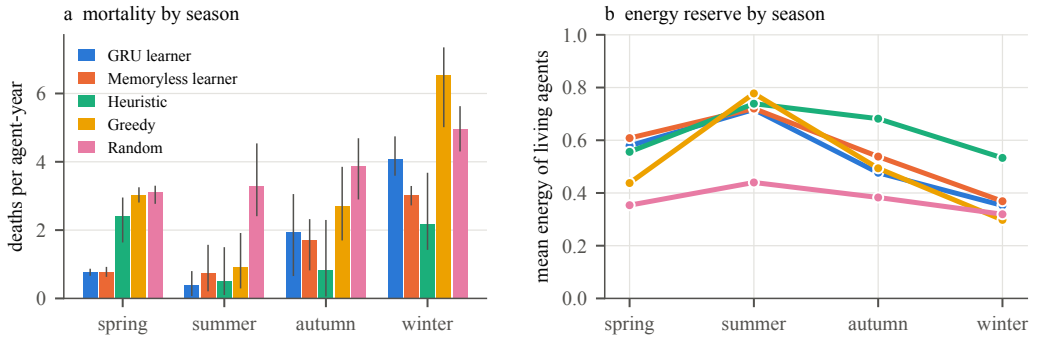


Figure 11: Mortality and energy by season (held-out worlds 0–4, 2 yr; bars are means, whiskers the range over worlds). (a) Learners are the best policies in spring and summer and the worst non-random policies in winter. The heuristic dies most in spring, when post-winter thirst catches it. (b) The learners enter winter with lower reserves (0.35–0.37) than the heuristic (0.53).

6.2 Memory: fewer thirst deaths, more starvation (H3 mixed)

GRU agents suffered 2.7× fewer thirst deaths than memoryless agents ($g = -1.15$), in the predicted direction. The cause-specific CIs touch, however, so by our rule the difference is not distinguishable at $n = 5$. All-cause mortality was *higher* with memory: the memoryless agent did better on 5/5 worlds ($g = +0.59$). H3 is therefore *mixed*. The thirst prediction holds directionally, and the "no worse overall" clause fails at this budget. Section 6.4 offers a mechanism.

6.3 The deficit is a winter deficit (exploratory)

Resolving deaths by season (Fig. 11) shows that the learners are not uniformly mediocre:

- **Spring and summer:** the learners are the *best* policies, at 0.38–0.78 deaths / agent-yr, against 0.50–2.42 for the heuristic.
- **Winter:** the GRU learner reaches 4.09 and the memoryless learner 3.03, against 2.16 for the heuristic.
- **Energy reserves** fall to 0.35–0.37 in winter, against 0.53 for the heuristic.

A shorter evaluation confirms this directly (Fig. 12). We re-ran all five policies on worlds 0–1 for half a year, which covers only spring and summer. Over that horizon both learners beat the heuristic: 0.19 and 0.33 against 1.22. Over two years they do not.

This has a methodological consequence. **Evaluations that do not span a winter would have re-**

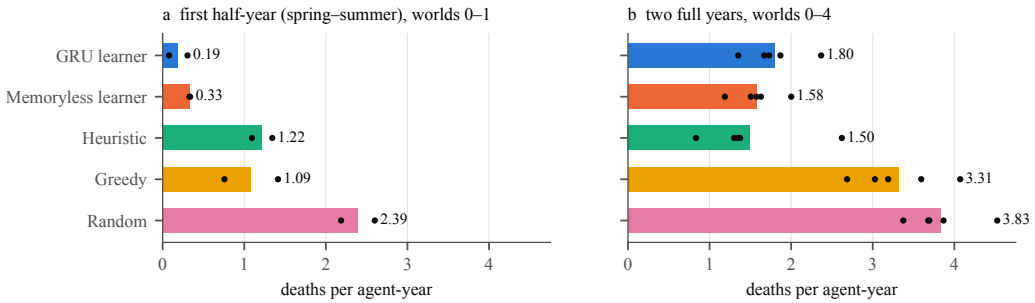


Figure 12: Evaluation horizon changes the ranking (exploratory). (a) Over the first half-year after spin-up (spring–summer; worlds 0–1, the protocol used by the in-training evaluations), learners have 4–6× lower mortality than the heuristic. (b) Over two full years they do not. Dots are individual worlds.

ported, wrongly, that the learners beat the hand-written rules. The in-training mini-evaluations of our replication runs use 0.5-year horizons. They are useful for tracking progress but must not be used for claims.

A natural hypothesis, which we have not yet tested, is that the effective planning horizon is too short. With $\gamma = 0.995$, credit decays over roughly 200 ticks (1.7 days), while a winter lasts 960 ticks (8 days). Storing energy in autumn pays off only beyond the discount horizon, which the hand-written "eat until 0.9" rule handles by fiat.

6.4 Energy economics (exploratory; relevant to preregistered H5)

We computed the actual energy gained per meal from the recorded events (each EAT event stores the energy gained) for all 9.8M learner agent-ticks per arm. Table 5 gives the result.

- **Intake** is 1.31×10^{-3} (GRU) and 1.34×10^{-3} (memoryless) per agent-tick, almost exactly basal metabolism (1.3×10^{-3}).
- **Plants are depleted.** Plants supply 73–74% of intake, but a plant bite yields only 0.0094 energy on average, less than a third of the 0.03 nominal value, because cells are grazed down.
- **Other sources** are fruit (23–25%) and meat (2–3%). Nearly all of the meat is scavenged: the GRU agents ate 2,692 meat bites but made only 25 small-prey kills and *no* large-prey kills in 2,560 agent-years. This is consistent with the preregistered H5 for the replication (at most 1 large kill per 1,000 agent-years).

A first-order budget adds basal cost, halved while resting, and 9×10^{-4} per move, scaled by terrain, so the move cost is a lower bound. It explains the arm difference:

- **GRU agents** move on 37% of ticks and rest on 16%. Their expenditure is $\gtrsim 1.52 \times 10^{-3}$, a deficit of about 2×10^{-4} per tick.
- **Memoryless agents** move on 24% of ticks and rest on 28%. Their expenditure is $\gtrsim 1.34 \times 10^{-3}$, roughly in balance.

In winter the memoryless agent rests on 48% of ticks, against 23% for the GRU agent (Fig. 13a). The better-surviving learner is the one that *does less*. One plausible reading is that the GRU agent explores more because its memory makes exploring for water pay off. That would explain both its lower thirst and its higher starvation, but we have not established it causally.

6.5 What the learners do

Policy entropy falls from 2.40 nats (uniform over 11 actions) to about 1.2–1.3 (Fig. 14a).

Table 5: Energy budget per agent-tick from recorded events (held-out worlds 0–4 × 2 yr). Expenditure is a first-order lower bound: basal 1.3×10^{-3}, halved when resting; moves 9×10^{-4} times the share of move actions; terrain, snow, body-size and carrying multipliers are omitted.

Arm	plant	fruit	meat	intake	move share	rest share	expenditure (≥)
GRU	0.965	0.297	0.044	1.306	0.365	0.162	1.52
Memoryless	0.981	0.330	0.032	1.342	0.243	0.276	1.34

Energy values $\times 10^{-3}$ per agent-tick.

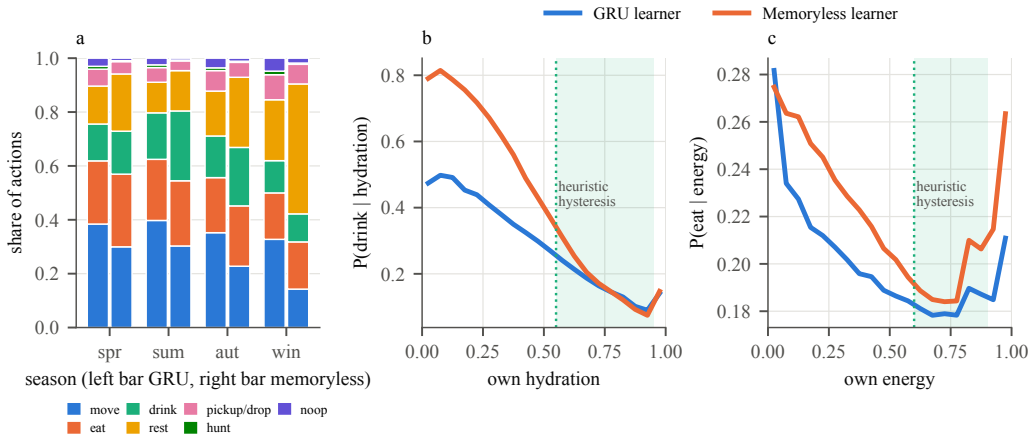


Figure 13: Learned behaviour (all agents, held-out worlds 0–4 × 2 yr). (a) Action budget by season. The memoryless agent shifts to resting in winter; the GRU agent keeps moving. (b, c) Learned "control curves": the probability of drinking or eating as a function of the agent's own need. The shaded band is the heuristic's hysteresis window. Both learners discovered a graded, need-dependent policy without being told the variables' meaning. The memoryless agent drinks more urgently when dehydrated.

- **Action budget** (GRU): 37% moves, 21% eat, 16% rest, 15% drink, 7% pickup/drop and 1% hunt.
- **Hydration** averages 0.88 for GRU and 0.83 for memoryless agents while alive.
- **Control curves.** Both learners' probability of drinking falls smoothly with hydration (from 0.5–0.8 near empty to under 0.1 near full), and the probability of eating falls with energy (Fig. 13b,c). This is a graded version of the heuristic's threshold rules, learned from the reward alone.

Behavioural inflexibility under drought (exploratory). While watching the viewer we noticed an agent repeatedly "drinking" at a stream that a drought had dried up. We quantified this from the recordings. The kernel logs a drink event whenever fresh water or floodwater is adjacent, or it is raining, so a DRINK action with no event means no water was in reach.

- **During droughts:** 78% (GRU) and 91% (memoryless) of drink actions are futile.
- **Outside droughts:** 30% and 37%.

Learned policies key drinking to places and body state, but do not fully condition on whether water is actually present. Droughts are rare in training, so this is a natural target for the continual-learning and world-model arms.

6.6 Training dynamics and the replication in progress

Training-time mortality is dominated by the world schedule and the seasons (Fig. 14b), so held-out evaluation is the only valid learning signal. The preregistered replication trains 3 seeds per arm for

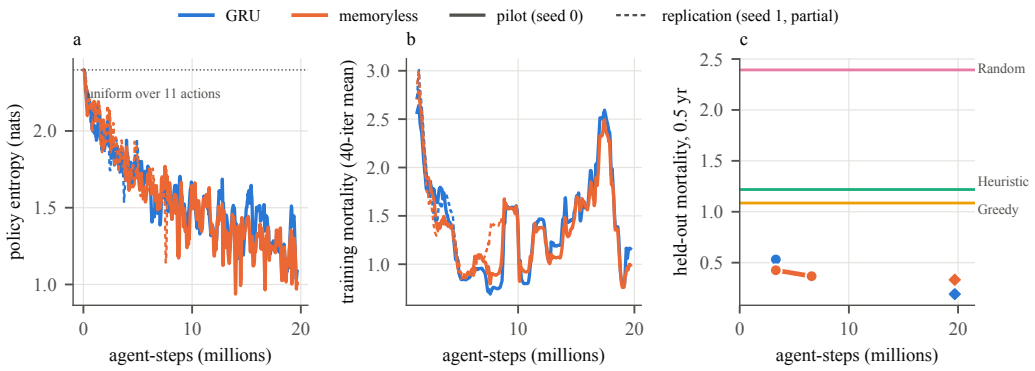


Figure 14: Training. (a) Policy entropy. (b) Training-time mortality (40-iteration moving mean). It oscillates with the 40-iteration world switch and the seasons and is *not* a usable learning curve. (c) Held-out mini-evaluations (worlds 0–1, 0.5 yr) of the partial replication runs (circles), the pilots' final checkpoints under the same protocol (diamonds), and matched baseline levels (lines). Note the half-year caveat of Fig. 12.

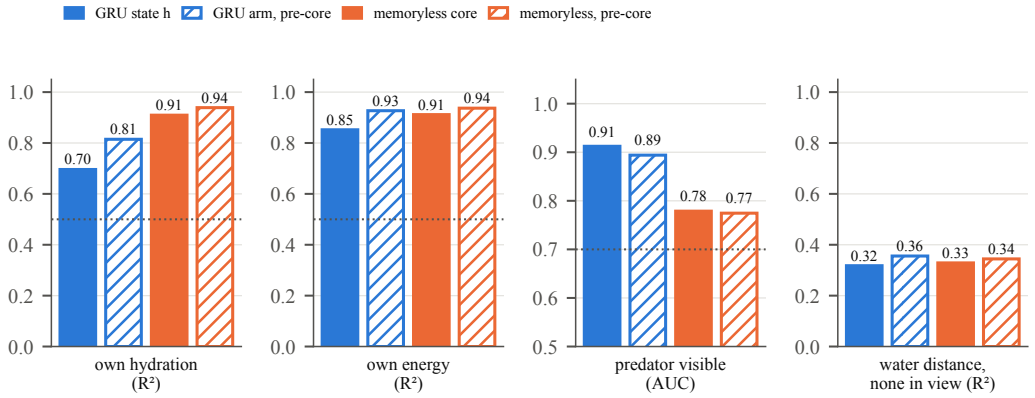


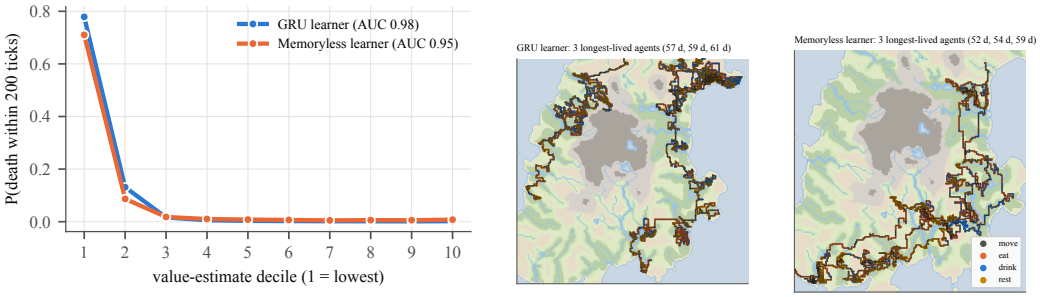
Figure 15: Linear probes (held-out world 0, first year, 122,880 samples per arm; train/test split by agent; dotted lines are the preregistered thresholds). Solid bars use the core output (GRU state h, or the memoryless core); hatched bars use the same network's pre-core encoding. The right panel is the memory test: decoding log distance to the nearest fresh water on ticks with *no water in view*. The GRU state does no better than the current encoding.

900 iterations (29.5M agent-steps) and evaluates every 100 iterations. It is *paused*, at 154/900 (GRU, seed 1) and 262/900 (memoryless, seed 1). Its first held-out points (0.53 and 0.43 at 3.3M steps; 0.37 at 6.6M for memoryless) fall well below the half-year baselines, consistent with the pilots (Fig. 14c). Its final two-year evaluation will decide H2′ and H3′. Separately, an independent re-run of the pilot protocol with the unified learner (PPOTrainer) reproduced the pilots within their CIs (GRU 1.67, memoryless 1.50 deaths / agent-yr).

6.7 Internal states, value and social structure (H6, H7 on pilot checkpoints)

H6 and H7 were preregistered for the replication runs. Here we apply the same analyses, with the same thresholds, to the pilot checkpoints and their evaluation recordings, so that the pipeline is tested before the decisive data arrive. The replication remains the confirmatory test. All numbers are in `stats_h6_h7.json`.

Probes (H6). We replayed held-out world 0 bit-exactly and re-ran each network on the reconstructed observations. Recomputed values match the recorded ones with correlation 1.000 and a



(a) value calibration

(b) the longest-lived agents on world 0

Figure 16: What the critic knows, and where agents go. (a) Probability of death within 200 ticks by decile of the recorded value estimate (worlds 0–4, about 2.4M samples per arm). The lowest decile dies 71–78% of the time, against a base rate of 9%. (b) The three longest-lived agents per arm (52–61 days), coloured by action. Ranges cover large parts of the island along river corridors.

maximum absolute difference of 1.3×10^{-6}. We then fit ridge and logistic probes with a train/test split by agent (118 test agents).

- **GRU state:** hydration $R^2 = 0.70$, energy $R^2 = 0.85$, predator-visible AUC = 0.91, against the preregistered thresholds of 0.5 and 0.7.
- **Memoryless core:** 0.91, 0.91 and 0.78.

H6 is supported on the pilots, with two caveats. First, hydration and energy are direct interoceptive inputs, and the pre-core encodings decode them even better (0.81–0.94), so this is a weak test of anything "internal". Second, a predator is visible on only 0.2–0.4% of samples. Label-shuffled controls give $R^2 \approx 0$.

The exploratory memory test asks something sharper. On ticks with no water in view (32–38% of samples), can the representation predict how far away the nearest water is? The GRU state reaches $R^2 = 0.32$, which is *lower* than its own pre-core encoding (0.36) and no better than the memoryless network (0.33). **We find no evidence that the GRU state stores where water is.** Its lower thirst mortality must therefore come from another mechanism, for example habitually staying closer to rivers (the GRU agents' mean distance to water is 3.4–4.2 cells, against 4.0–4.7 for the heuristic).

Value calibration (exploratory). Across all five held-out worlds, a low recorded value estimate predicts death within 200 ticks with AUC 0.98 (GRU) and 0.95 (memoryless) (Fig. 16a). The critic has learned an accurate model of its own mortality risk. This is a functional correlate useful for analysis, not evidence of anything felt.

Cooperative hunting (H7, part 1). The learners attacked large prey 14 times (GRU) and 7 times (memoryless) in 2,560 agent-years each, killing none.

- **GRU:** one attack episode involved two agents within 6 ticks, against 0.34 expected under the shuffled-timing null ($p = 0.34$).
- **Memoryless:** none.

The null part of H7 holds, but with almost no statistical power. The baselines teach a methodological lesson. Random and heuristic agents, which have no social rules at all, "beat" the same null: 2 multi-attacker episodes each, $p = 0.004$ and $p = 0.04$. Agents drawn to the same herd coincide in time more than shuffled timing allows. **A timing-only null is therefore anti-conservative for coordination**, and the replication analysis will use a null that also preserves spatial co-location, calibrated on baseline agents.

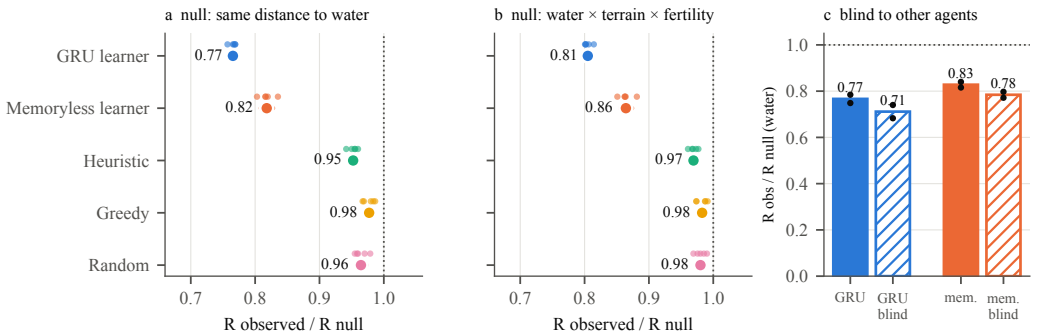


Figure 17: Aggregation and its cause. (a, b) Clark–Evans R relative to two nulls (mean and 95% CI over worlds 0–4; small dots are worlds; < 1 means more clustered than the null). Learners cluster beyond water and habitat, while the baselines do not. (c) Blinding the trained policies to other agents (agent channel and neighbour features zeroed; worlds 0–1, 1 yr) does not remove the clustering; it increases it. The clustering is not mediated by perceiving others.

Aggregation (H7, part 2). Fig. 17 compares the Clark–Evans index R (Clark and Evans, 1954) of the live agents, sampled daily on worlds 0–4 over 2 years, with two nulls. The first null relocates each agent to a random passable cell at the same distance from fresh water. The second also matches terrain class and fertility tercile.

- **Baselines** sit close to their nulls: ratios 0.95–0.98.
- **Learners** are markedly more clustered: 0.77 [0.76, 0.77] (GRU) and 0.82 [0.81, 0.83] (memoryless) under the water null, and 0.81 and 0.86 under the habitat null.

So the second clause of H7, that aggregation does not differ from the heuristic once water is controlled, is **not supported**.

Is the clustering social? We tested this causally. We re-ran both trained policies on worlds 0–1 with the agent channel of the grid and all neighbour features set to zero, which makes them *blind to other agents*. Clustering did not weaken; it *strengthened*, from 0.77 to 0.71 (GRU) and from 0.83 to 0.78 (memoryless). Mortality was unchanged: 1.74 vs. 1.73 and 1.49 vs. 1.52 deaths / agent-yr. The learners therefore cluster *despite* seeing each other, not *because* of it; if anything, seeing others slightly disperses them. Two non-social explanations remain open: a shared policy responding identically to dynamic resources that the static nulls ignore (grazed plant patches, fruiting trees), and synchronised resting sites. **We do not count this clustering as emergent social structure.** Without the ablation, it would have been easy to misreport it as such.

6.8 Population density (Stage-1 finding)

Water-seeking baselines collapse at high density. At 1,000 agents on the same map, the heuristic reaches 4.94 deaths / agent-yr, with 96% of deaths from thirst, which is *worse* than random (3.85). At 256 agents it reaches 1.72. Our hypothesis is congestion at water, which we have not yet tested. Learners must therefore always be compared with baselines at the same population size.

7 Measuring emergence: framework and preregistered nulls

The point of Genesis is to measure social structure without having put it there. Table 6 lists the Stage-3 instruments and Fig. 18 the pipeline. Each has three parts:

1. an operational definition computable from the flight recorder;
2. a null model that removes the social explanation while preserving ecological confounds;

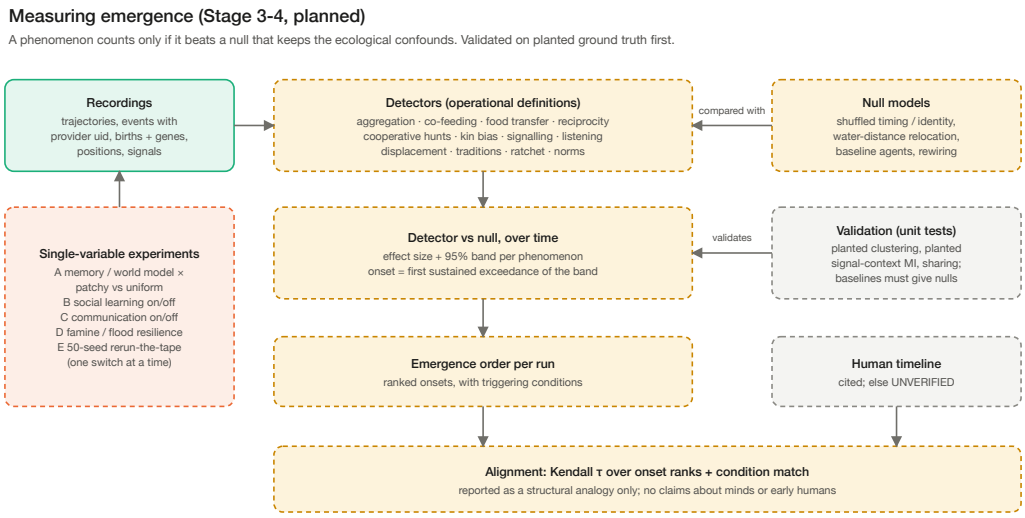


Figure 18: Measuring emergence (planned). Detectors are computed from the recordings and compared with null models that keep ecological confounds; they are validated on planted ground truth. Onsets give an emergence order, which is compared with a cited human timeline as a structural analogy only.

3. a unit test on synthetic data with planted ground truth.

The onset of a phenomenon is the first sustained exceedance of its null's 95% band. Its *order* across phenomena is what will eventually be compared, as a structural analogy only, with a cited human timeline in which every entry is verified or flagged `UNVERIFIED`. The comparison uses Kendall's τ over onset ranks.

Preregistered for the replication (H5–H7).

- **H5:** starvation stays the leading cause, with at most 1 large-prey kill per 1,000 agent-years.
- **H6:** a linear probe on the GRU state decodes hydration and energy ($R^2 \geq 0.5$) and predator presence (AUC $\geq$ 0.7). Such results would be functional correlates only, never claims about feelings.
- **H7:** multi-attacker large-prey attacks do not exceed a shuffled-timing null, and aggregation does not differ from the heuristic once distance to water is controlled.

Pilot-checkpoint results for H6 and H7 are in §6.7: H6 is met, cooperative hunting is null, and the aggregation clause fails for a non-social reason. We expected H7 to hold, a null, in the capability-off setting. Social structure, if any appears, should require the switches of Table 1.

8 Discussion

What the pilots show. The survival-only learners are competent in the sense that they beat naive foragers and match a hand-designed controller. Their competence has a specific shape:

- they handle water, predators and fire better than our rules do;
- they handle the energy economy, and especially winter, worse.

The world, not the learner, sets this ceiling. Plants are grazed down to about a third of their nominal value, prey are hard to catch, and large prey require coordination that the agents do not achieve. We will not tune the world to rescue the learner. If no strategy can generate a surplus, that is itself

Table 6: Stage-3 measurement framework (built: ✓; designed and preregistered but not yet run: ○). All social metrics are reported against both a baseline-agent null and a shuffled null.

Phenomenon	Operational metric	Null / control	
Aggregation	Clark–Evans R (Clark and Evans, 1954); group-size distribution	water- and habitat-matched relocation; social blinding	✓
Co-feeding	meals from carcasses killed by others (provider uid)	carcass-density null	✓ (count)
Food transfer	DROP by $A \rightarrow$ EAT/PICKUP by B within k ticks	identity shuffle	○
Reciprocity	corr. of $A \rightarrow B$ and $B \rightarrow A$ transfers	degree-preserving rewiring	○
Cooperative hunt	large-prey attacks with ≥ 2 attackers in 6 ticks	shuffled timing (anti-conservative; §6.7)	✓
Kin bias	transfers/proximity vs. genetic similarity	gene-label permutation	○
Signalling	I(signal; context) (positive signalling)	shuffled contexts	○
Listening	causal effect of a heard symbol on action (Lowe et al., 2019)	symbol intervention	○
Displacement	signals predicting out-of-view resources (Hockett, 1960)	in-view-only control	○
Traditions	group-specific behaviours not explained by ecology/genes (Whiten et al., 1999)	ecology-matched groups	○
Ratchet	gen.-n performance with vs. without social learning (Tennie et al., 2009)	social learning off	○
Norms	strikes conditional on target's sharing history	random targeting	○
Resilience	mortality impact and recovery time after disturbances	matched undisturbed periods	○
Internal states	linear probes (Alain and Bengio, 2016) for hydration, energy, predator	pre-core encoding; shuffled labels	✓
Value calibration	AUC of value for death within 200 ticks	base rate	✓

a property of the ecology to report. The more interesting question is whether the *only* route to a surplus is cooperative. Two candidates are hunting large prey together and caching food in autumn, which are exactly the affordances the testbed exposes.

Look-alikes of social structure. The pilots already produced two phenomena that a less careful pipeline would have reported as emergent sociality. First, learners cluster far beyond what water and habitat explain. Second, baseline agents with no social rules "coordinate" large-prey attacks beyond a timing null. Neither survives the right control: social blinding for the first, a baseline-calibrated null for the second. We take this as the main methodological lesson so far. Every candidate social phenomenon needs a causal ablation of the social channel and a null calibrated on agents that provably lack the capacity.

Why this matters for emergence claims. If cooperation later appears, we need to know whether it appeared because it was the only way out of an energy trap or because of a capability switch. Seasonal, cause-resolved and energy-resolved accounting of the kind in §6 is therefore part of the measurement apparatus, not a side analysis. The horizon result (Fig. 12) shows how easily a protocol choice can reverse a conclusion.

Limitations.

- **Pilot scale.** There is one training seed per arm, so the variance between training runs is unmeasured. The 3-seed replication is paused part-way.
- **Budget.** 19.7M agent-steps is small for a 1.1M-parameter recurrent policy, and memory benefits may need more training.
- **Determinism.** MPS arithmetic is not bit-reproducible. Worlds, sampling and the CPU path are.
- **Learner versions.** The pilots were trained with the original `agents/rppo.py`. It has since been folded into `agents/ppo.py` (PPOTrainer), whose defaults are now the pilots' settings. An independent re-run with PPOTrainer reproduces the pilots within their CIs: 1.67 [1.45, 1.92] (GRU) and 1.50 [1.30, 1.75] (memoryless) deaths / agent-yr (`reports/stage2/STAGE2_REPORT.md`). The pilot checkpoints predate the current checkpoint format and are loaded by `pilot_net.py` in this directory.
- **Exploratory analyses.** The seasonal, energy, horizon and drought analyses were not preregistered. The half-year comparison uses only two worlds. H6 and H7 were preregistered for the replication and are applied here to pilot checkpoints. Probes use one world, and the blinding ablation uses two worlds and one year.
- **Ecology.** Predators hit their immigration floor about 31% of the time under heuristic pressure.
- **Scale of the world.** It is simple compared with any real ecology, and all comparisons to human prehistory are structural analogies.

Broader impact. GENESIS studies how self-interested learners could come to cooperate or conflict, a question relevant to AI safety and to multi-agent deployment. The system runs on a single laptop, and every run is recorded and exactly replayable, which supports auditing. We see no direct misuse pathway. The main risk is over-interpretation, such as reading agent behaviour as evidence about human minds. We address this through restrained language, preregistration and baselines.

9 Roadmap and conclusion

GENESIS is designed to be run as a sequence of single-variable experiments:

- **A.** Baseline vs. memory vs. world-model agents (a Dreamer-style latent model (Hafner et al., 2023)), crossed with patchy vs. uniform resources.
- **B.** Social learning on or off across generations: does a cultural ratchet appear?
- **C.** Communication on or off: does referential or displaced signalling appear, and do dialects drift?
- **D.** Resilience to famine and floods across A–C.
- **E.** A 50-seed rerun-the-tape study at reduced scale.

Continual-learning probes (Dohare et al., 2024; Kirkpatrick et al., 2017) and lifelong agents with parents and young come next.

The first study already shows what such a programme needs: held-out worlds, horizons that include the hard season, non-learning baselines, cause-resolved accounting and preregistered decisions. Under those conditions, survival-only learners turn out to be good at water, bad at winter, and as yet no better than a page of hand-written rules. We will report whether anything social emerges with the same rigour, including if the answer is no.

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Appendix

A World details

Table 7: Terrain classes.

Class	Passable	Move cost	Notes
Deep water	no	–	ocean is salt; deep lake cores fresh (drink from shore)
Shallow	yes	×2	rivers, streams, lake rims fresh; ocean shoreline salt
Sand, grass, meadow, barren	yes	×1	fertility differs (meadow highest)
Forest	yes	×1.25	wood, most fruit trees, +2 °C, two cells block sight
Rock	yes	×1.6	stone; hills
Mountain	no	–	blocks line of sight
Cave	yes	×1	shelter: +10 °C, rain cover, predators never enter

Table 8: Key ecological parameters (`configs/default.yaml`).

Plant growth rate	0.0045 /tick	Plant bite / energy per biomass	0.25 / 0.12
Season growth (Sp/Su/Au/Wi)	1.2 / 1.0 / 0.55 / 0.05	Grazing floor	10% of capacity
Fruit energy / max per tree	0.15 / 6	Meat bite / energy per unit	0.2 / 1.0
Drink amount / rain drink	0.22 / 0.006	Hydration cost	0.0016 /tick
Basal / move cost	0.0013 / 0.0009	Rest factor (basal)	0.5
Small prey: kill p / carcass	0.6 / 0.30	Large prey: attackers / window	$\geq$ 2 / 6 ticks
Predator: attack p / damage	0.4 / 0.08	Fauna torpor below 5 °C	metabolism ×0.2
Fire ignition / spread	0.04 /day / 0.22	Famine probability	0.08 /year
Drought (summer)	0.06 /day, 6–14 d	Cold snap (winter)	0.05 /day, 3–7 d, −10 °C

B Observation and network details

The 13 self features are energy, hydration, health, warmth, age fraction, carried food, carried wood, carried stone, felt temperature, light, raining, sheltered and maturity. Each of the 4 nearest visible neighbours contributes 9 features: present, dx, dy, health, carried food, last action, share rate, strike rate and kin similarity. Last action is zero unless `social_learning` or `reputation` is on; share and strike rates are zero unless `reputation` is on. The grid is cast to float and divided by 255. The GRU and memoryless arms differ only in the core layer.

C Engineering for exact reproducibility

- **Bit-identical runs.** `test_full_run_bit_reproducible` asserts that two runs with the same seed produce identical trajectory hashes, and that different seeds differ.
- **Sequential/parallel twins.** Kernels that run sequentially for small workloads (macOS thread launch costs about 60 μs) have parallel twins whose outputs are tested to be identical.
- **Flight recorder.** `sim/recorder.py` stores, for every agent and every tick, the state the agent acted from, together with its action, reward, death and cause. It also stores learner internals (value, action probabilities, log-probability), every animal's state, world totals, events, births with genes, and every manual intervention.
- **Replay.** `analysis.trajectories.replay` re-simulates a recording and checks every agent at every tick bit-exactly, so observations need not be stored (about 2 KB per agent per tick otherwise).

- **Size.** About 63 MB per simulated year at 256 agents.

The suite has 55 tests, including reward purity, world rules, the PettingZoo API, observation-image drift, twin agreement, replay and a deterministic end-to-end training loop.

D Live viewer

A Python aiohttp server runs the simulator in a background thread and streams binary frames over a WebSocket. A three.js client renders a perspective "diorama" of the world, with weather, seasons, a day/night cycle, articulated agents, an inspector showing a selected learner's value estimate and action probabilities, overlays and a measured-only dashboard. Experiment controls (speed, fire, flood, drought, cold snap, famine, restart) are applied between ticks and logged, so speed never changes results. Sessions longer than one simulated day append a summary to the agent log.

E Per-world results

Table 9: Deaths per agent-year on each held-out world (2 yr, 256 agents).

Policy	world 0	world 1	world 2	world 3	world 4
Random	3.69	3.87	4.53	3.37	3.68
Greedy	3.59	3.19	4.07	3.03	2.69
Heuristic	1.37	1.35	2.62	0.83	1.30
GRU learner	1.73	1.87	2.37	1.35	1.67
Memoryless learner	1.63	1.58	2.00	1.19	1.51

Table 10: Mortality by season (deaths / agent-yr, mean over worlds 0–4) and **mean energy of living agents**.

	mortality				energy			
Policy	Sp	Su	Au	Wi	Sp	Su	Au	Wi
GRU	0.76	0.38	1.93	4.09	0.58	0.72	0.48	0.35
Memoryless	0.78	0.73	1.71	3.03	0.61	0.72	0.54	0.37
Heuristic	2.42	0.50	0.83	2.16	0.56	0.74	0.68	0.53
Greedy	3.03	0.92	2.70	6.55	0.44	0.78	0.49	0.30
Random	3.10	3.28	3.87	4.97	0.35	0.44	0.38	0.32

Table 11: Population-density sweep (baselines, 3 seeds × 1 yr): deaths / agent-yr (thirst share).

Policy	256 agents	512 agents	1,000 agents
Random	3.61 (0.38)	3.25 (0.33)	3.85 (0.33)
Greedy	3.61 (0.49)	3.91 (0.71)	5.50 (0.78)
Heuristic	1.72 (0.56)	1.64 (0.86)	4.94 (0.96)

F Reproducibility

- **Code.** https://github.com/HeetRanpura/Genesis. Pilot checkpoints were trained at commits `c5ea9ff` (GRU) and `50143db` (memoryless) and evaluated at `e1d919e`. The analysis scripts in `reports/stage2/pilot_preprint/` run on current `main`.

- **Software.** Python 3.11, PyTorch 2.14.1 (MPS), Numba 0.68.0, NumPy 2.4.6.
- **Hardware.** Learners ran on a MacBook Pro (M3 Pro, 11 cores, 18 GB). Engine benchmarks ran on a fanless MacBook Air (M4). Throughput is never compared across machines.
- **Seeds.** Training worlds are seeds 1000+ (one new world every 40 iterations). Evaluation worlds are 0–4. The training seeds are 0 (pilots) and 1–3 (replication).

```
make setup && make test                      # environment + tests
.venv/bin/python experiments/run_baselines.py --seeds 5 --years 2
# pilots: trained/evaluated with the original rppo scripts (c5ea9ff, 50143db,
#         e1d919e); today's learner: make train NAME=gru_s0 / make eval NAME=...
.venv/bin/python reports/stage2/pilot_preprint/make_figures.py   # every figure + stats.json
.venv/bin/python reports/stage2/pilot_preprint/analysis_h6_h7.py  # probes, value, hunts, paths
.venv/bin/python reports/stage2/pilot_preprint/aggregation_strict.py  # water + habitat nulls
.venv/bin/python reports/stage2/pilot_preprint/blind_ablation.py  # social blinding
make viewer POLICY=ppo:gru_s0                # live viewer, http://localhost:8787
```

- **Run directories.** Each run directory contains the resolved config, a manifest (git commit, versions), `metrics.csv`, `events.npy` and a `trajectory/` directory of compressed 480-tick chunks with a start-state hash. Replay of an older recording fails loudly if the engine's dynamics have changed since.
- **Analysis outputs.** `stats.json` holds every number quoted in §6.

G Checklist (abridged NeurIPS)

- **Claims match results?** Yes. Negative results (H2, H3) are stated in the abstract.
- **Limitations discussed?** Yes, in §8.
- **Code and data available?** Yes, the repository; run data (several GB) is available on request.
- **Training details?** Yes, in Table 3 and Algorithm 1.
- **Error bars?** Yes: 95% bootstrap CIs over worlds, plus Hedges' g. The unit of replication is stated.
- **Compute?** About 50 minutes per pilot run on an M3 Pro (MPS), and about 45 s per 2-year evaluation world.
- **Preregistration?** Yes, a timestamped file committed before evaluation. Exploratory analyses are labelled as such.
- **Human subjects or sensitive data?** None.

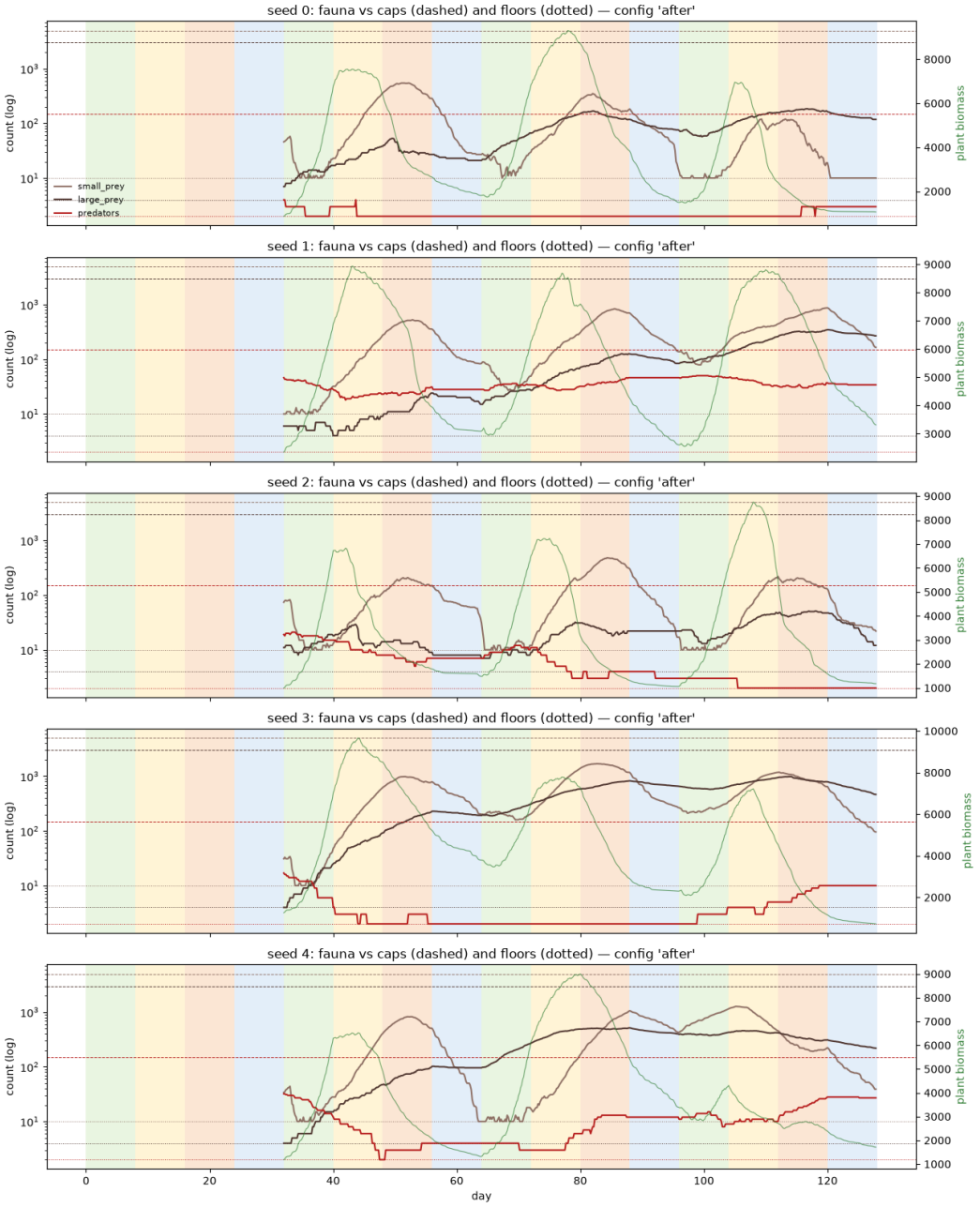


Figure 19: Food-limited fauna (Stage 1.1). Populations are regulated by food, torpor and seasonal breeding, not by caps, which bind 0% of the time.

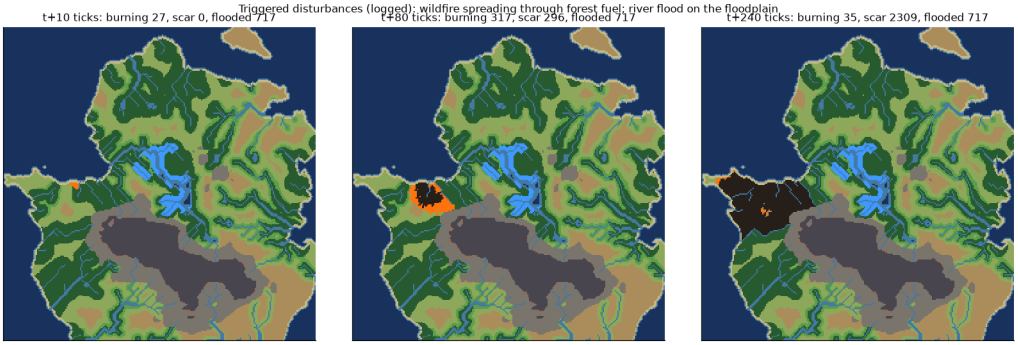


Figure 20: Disturbances: wildfire spread and burn scars, spring floods, drought-dried streams and snow.

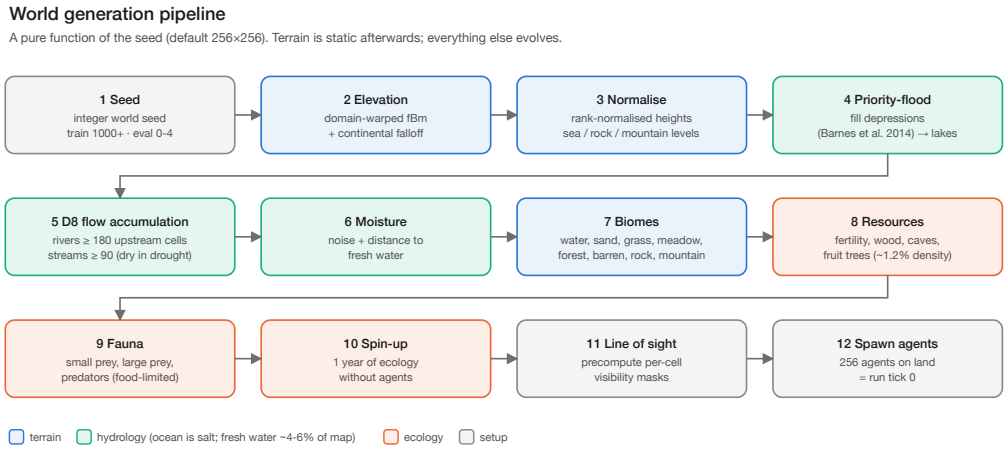


Figure 21: World generation. The world is a pure function of the seed. Terrain is static afterwards, which is what allows the line-of-sight masks to be precomputed.

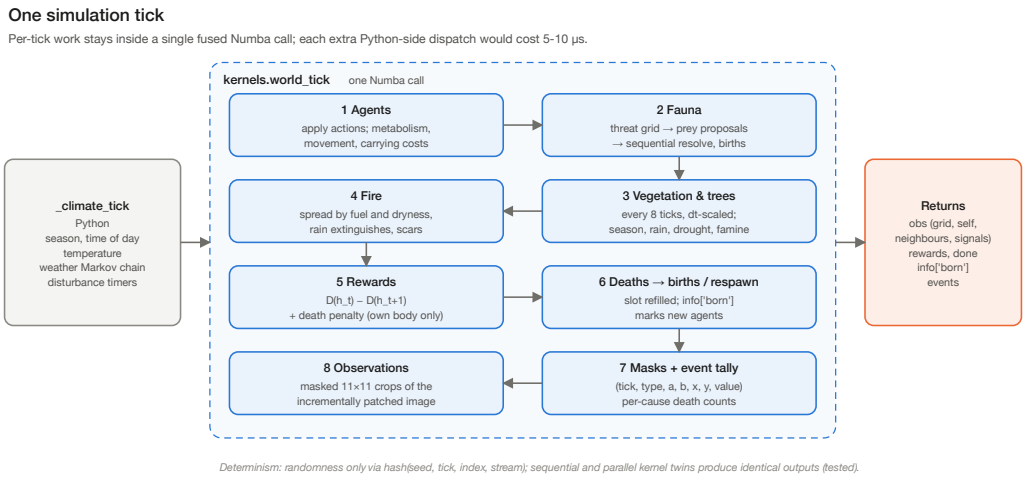


Figure 22: One simulation tick. A Python climate update followed by one fused Numba call that performs all per-tick work in a fixed order.

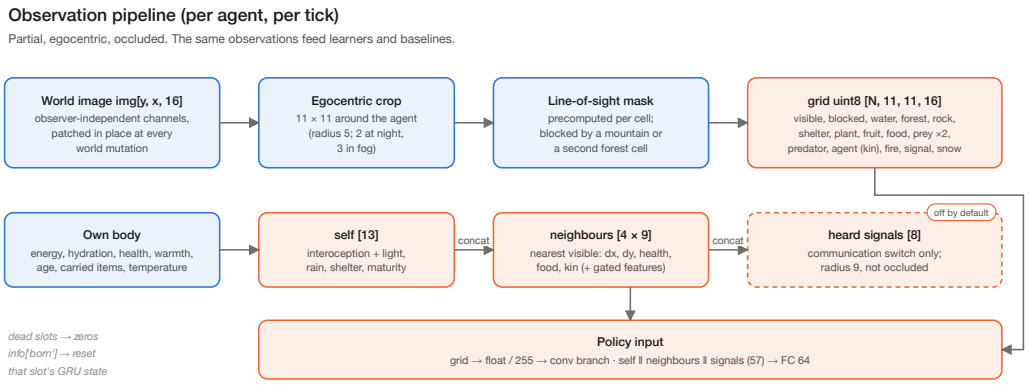
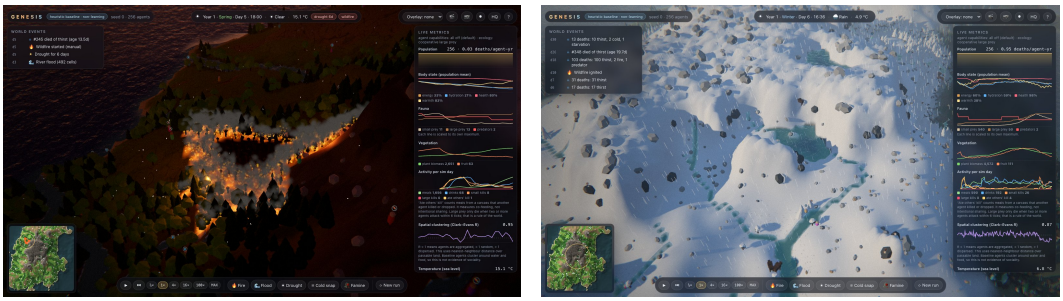


Figure 23: Observation pipeline. Crops of an incrementally maintained world image are masked by precomputed line of sight and combined with interoceptive and neighbour vectors.



(a) triggered wildfire **(b)** winter



(c) inspector following one agent

Figure 24: Viewer screenshots. Disturbance controls trigger the same logged `trigger_*` methods the experiments use.